

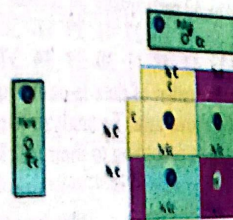
UNIT 11

BASIC STATISTICS

In this unit the students will be able to:

- Construct grouped frequency table, Histogram(unequal intervals) and frequency polygon.
- Calculate Arithmetic mean, Median, Mode of grouped data.
- Calculate weighted mean.
- Related real life word problems
- Probability of single events.
- Probability of complementary events.
- Relative frequency as an estimate of probability.
- Expected frequency.
- Related real life word problems.

The famous Biologist Gregor Mendel demonstrated that pea plant characteristics were transmitted as discrete units from parent to offspring i.e. in a genetic cross, the probability of a pea plant of purple flower with a pea plant of yellow flowers, the dominant trait is expressed in 3 out of 4 of the offspring and recessive trait is expressed in one out of 4 as shown in the adjoining figure.



Statistic plays vital role in our day to day life as in our lives we have to come across frequent decisions based on some relevant past experiences or the related statistics already available. For instance we may use statistics in efficient budgeting of our family, in finding the best prices for sale or purchase of products of our interest, to predict risks in our business, to estimate best time to perform various routine activities. Online shopping is an easily understandable example of this era where the customers see reviews of others about a product and make decision to purchase or not to purchase. If more than 50% reviews are in the favour it may be predicted that the product will be out of stock soon. Similarly, to estimate the chances of failure or success of a project, probability has great application. Quantifying and estimating investment and risk assessment in a project are very important for financial planning, investment comparison and decision making.

THINK MORE

Hadiya is on the holy trip of umrah. While her stay she observed the best time to perform TWWAF at the KAABAH PAK when the crowd is smallest soon after Fajar prayer when most of the people go back to rest. What time do you think it might be next.



11.1 Frequency Distribution

In statistics it is very important to represent a data in a manageable way so that it becomes easy to understand, analyze and to draw useful results. This useful presentation of data can be made by various methods. Grouped frequency distribution is one of the most important methods to represent the quantitative data. Data can be classified as:

Raw Data / Ungrouped Data

Data which has not been arranged in a systematic order is called raw data or ungrouped data.

Grouped Data

A tabular arrangement of data in which various items / values are arranged or distributed into some classes and the number of items falling in each class is stated is called grouped data or a frequency distribution.

Let us try to understand the above two categories of data with the help of an example. The points scored by 50 participants of Maths Olympiad out of 100 are recorded below.

10 18 38 15 30 27 41 39 16 50 50 45 34 37 34 09 12 45 19 23 29 31 28 17 08
19 25 39 42 48 11 30 21 34 37 50 46 08 15 15 16 21 28 32 37 41 19 29 48 50

This presentation of data is called ungrouped data. It is difficult to draw any useful conclusions from the above data. To analyze the above results easily, it is better to distribute the students in some groups according to their marks as follows.

Marks obtained	No. of students
1 – 10	4
11 – 20	12
21 – 30	11
31 – 40	11
41 – 50	12
Total = 50	

From the table it is easy to get the following results.

There are 4 participants whose scores are from 1 to 10.

There are 12 participants whose scores are from 11 to 20.

There are 11 participants whose scores are from 21 to 30.

There are 11 participants whose scores are from 31 to 40.

There are 12 participants whose scores are from 41 to 50.

34 participants scored more than 20 points.

Data represented in this form is called grouped data.

To convert an ungrouped data into grouped form, the students should be familiar with the following terms.

(i) Number of Groups

A number of groups depends upon the size of data. There is no hard and fast rule for a number of groups, but it is usually between 5 and 15.

(ii) Class Limits

A group or class is defined by two numbers. The smaller number is called lower class limit and larger one is called upper class limit. In above example first group is (1 – 10), 1 is its lower class limit and 10 the upper class limit. Both the lower and upper limits are included in the respective groups.

(iii) Class Boundaries

In above example the values / marks between 10 and 11 are not included in any group. If marks of a student are 10.5, then above mentioned classes are not suitable. To include such values we should write the classes as

0.5 – 10.5, 10.5 – 20.5, 20.5 – 30.5, 30.5 – 40.5, 40.5 – 50.5.

The participants having scores between 0.5 and 10.5 including 0.5 are considered in the first group. The participants having scores 10.5 will be in second group. The largest value of 0.5 – 10.5 and smallest value of 10.5 – 20.5 are same.

0.5 and 10.5 are lower and upper boundaries of first group. The class boundaries can be formed from class limits as follows. First of all find difference between the lower class limit of second class and upper class limit of first class. Then subtract half of this difference from lower limits of all classes and add it to the upper limits of all classes.

$$\text{Adjustment factor} = \frac{1}{2} (\text{lower limit of a class} - \text{upper limit of the preceding class}) = 0.5$$

To understand the difference between class limits and class boundaries, look at the following table.

scores (class intervals)	Lower limits	Upper limits	Lower boundaries	Upper boundaries	Class boundaries
1 – 10	1	10	$1 - 0.5 = 0.5$	$10 + 0.5 = 10.5$	0.5 – 10.5
11 – 20	11	20	$11 - 0.5 = 10.5$	$20 + 0.5 = 20.5$	10.5 – 20.5
21 – 30	21	30	$21 - 0.5 = 20.5$	$30 + 0.5 = 30.5$	20.5 – 30.5
31 – 40	31	40	$31 - 0.5 = 30.5$	$40 + 0.5 = 40.5$	30.5 – 40.5
41 – 50	41	50	$41 - 0.5 = 40.5$	$50 + 0.5 = 50.5$	40.5 – 50.5

(iv) Size of Class Interval

Size of a class interval is the difference of lower limits or upper limits of two consecutive classes. It is usually denoted by 'h'.
e.g. in 1 – 10, 11 – 20, 21 – 30, the size of class intervals is 10.

(v) Class Marks / Mid Values

It is the average of lower and upper limits or boundaries of a class. Class marks are denoted by x .
Class mark of 1 – 10 is $\frac{1+10}{2} = 5.5$ and that of 11 – 20 is $\frac{11+20}{2} = 15.5$

Note:

1. The difference of class marks of two consecutive classes is equal to the size of class interval h .
2. Class boundaries may be derived from the class marks x and size of class interval h by using the formula $(x \pm \frac{h}{2})$.

(vi) Frequency

Number of repetitions of an individual value in an ungrouped data is called its frequency. In a grouped data, number of values present in a group is called frequency of group. In Table 1 frequency of group 1 – 10 is 4, because there are four students whose obtained marks are between 1 and 10 both inclusive.

(vii) Cumulative Frequency

The sum of frequency of a class and the frequencies of all the preceding classes is called cumulative frequency.

10.1.1 Construction of Frequency Distribution Table

We can construct a frequency distribution table in two ways.

- (i) Tally bar method (ii) Direct observation method

Tally Bar Method

Step I: Decide how many groups should be made in the table.

Step II: Find approximate size of class interval by the formula $h \approx \frac{x_l - x_s}{n}$,

Where h = size of class interval

x_l = largest value in ungrouped data

x_s = smallest value in ungrouped data

n = approximate number of classes

Step III: Draw a table and write the class limits in first column according to the size of class interval.

Step IV: Choose the values from ungrouped data one by one and draw small tally marks in second column against the respective group. If 5 or more values fall in a same group, then draw 5th tally mark diagonally. It will make the counting of tally marks easier.

Step V: Count tally marks of each group and write their number against the respective class in third column (column of frequencies).

Step VI: Find sum of all frequencies and check that it must be equal to the number of values present in ungrouped data i.e
number of total items in ungrouped data (n) = sum of frequencies in grouped data (Σf)

Example 1:

The following are masses in pounds of 40 boys at a gym.

147, 138, 164, 150, 132, 144, 125, 149, 157, 146, 158, 140, 136, 148, 152, 144, 168, 126, 138, 176, 163, 126, 154, 165, 146, 173, 142, 147, 135, 153, 140, 135, 161, 145, 135, 142, 150, 156, 145, 128.

Make a frequency distribution table using an appropriate size of class interval by tally bar method.

Solution:
Step I:

Let us distribute the data in 5 classes.

Step II:

$$x_l = 176, x_s = 125, n = 40.$$

$$h = \frac{x_l - x_s}{n} = \frac{176 - 125}{5} = 10.2 \approx 11$$

Step III:

Masses in pounds	Tally marks	Frequencies
125 – 135		8
136 – 146		13
147 – 157		11
158 – 168		6
169 – 179		2
Total		40

History a Mystery

Tally sticks are the ancient tools used for recording numerical data. These elongated pieces of bone, wood or ivory were marked with counts or lines. Marco Polo have documented the use of tally sticks in regions like Yunnan (China).

Direct Observation Method

In this method the values of ungrouped data are written in second column of table against the respective group instead of tally marks. The remaining process of construction of frequency table by direct method is same as that of tally bar method. When ungrouped data are not arranged in any order, then it is better to use tally bar method.

In example 1, the frequency table by direct method can be constructed as follows.

Masses in pounds	Observation	Frequencies
125 – 135	132, 125, 126, 126, 135, 135, 135, 128	8
136 – 146	138, 144, 146, 140, 136, 144, 138, 146, 142, 140, 145, 142, 145	13
147 – 157	147, 150, 149, 157, 148, 152, 154, 147, 153, 150, 156	11
158 – 168	164, 158, 168, 163, 165, 161	6
169 – 179	176, 173	2
Total		40

Discrete Frequency Distribution

If data is not in fractions and some values occur most often to constitute the whole data, then data is called a discrete data there is no need of class boundaries and size of class interval is 1.

Example 2:

A cubical die is rolled 30 times and numbers of dots on front face recorded each time are as follows.

2, 6, 2, 3, 2, 4, 5, 4, 3, 1, 2, 6, 5, 3, 4, 6, 2, 2, 4, 6, 5, 1, 3, 4, 6, 5, 1, 1, 6, 3

Make a discrete frequency distribution. Also make a column of cumulative frequency

Solution:

No. of dots	Tally marks	Frequencies	Cumulative frequency
1		4	4
2		6	4+6=10
3		5	4+6+5=15
4		5	4+6+5+5=20
5		4	4+6+5+5+4=24
6		6	4+6+5+5+4+6=30
Total	—	30	

11.1.2 Construction of Histogram

Different types of charts and graphs are used to represent a data effectively. It becomes very easy to get a quick idea about the data and to analyze by means of graphs and charts. In histogram, data are represented by adjacent rectangles having bases along horizontal axis (marked off by class boundaries) and areas proportional to the class frequencies. If the class interval sizes is the same, the heights of rectangles are proportional to the class frequencies as you have done earlier in grade 8. If the class interval size is unequal, then the heights of rectangles are to be adjusted.

Construction of Histogram with Unequal Size of Class Intervals

If sizes of class intervals are unequal, then to make areas of rectangles of a histogram proportional to their frequencies, the heights of rectangles have to be adjusted. In a histogram area of rectangle is proportional to class frequency.

$$\text{Area} \propto \text{frequency} \quad \text{or} \quad (\text{height} \times \text{width}) \propto \text{frequency}$$

$$\Rightarrow \text{height} \propto \frac{\text{frequency}}{\text{width}}$$

Thus height of rectangle may be found by dividing the frequency by width of class interval. The ratio $\frac{\text{frequency}}{\text{width}}$ is called **Frequency Density. (F.D)**

To make a histogram with unequal sizes of class intervals, we plot class boundaries on x – axis (horizontal) and frequency densities on y – axis (vertical).

Example 3: The heights in cm of 44 students of a school is given below.

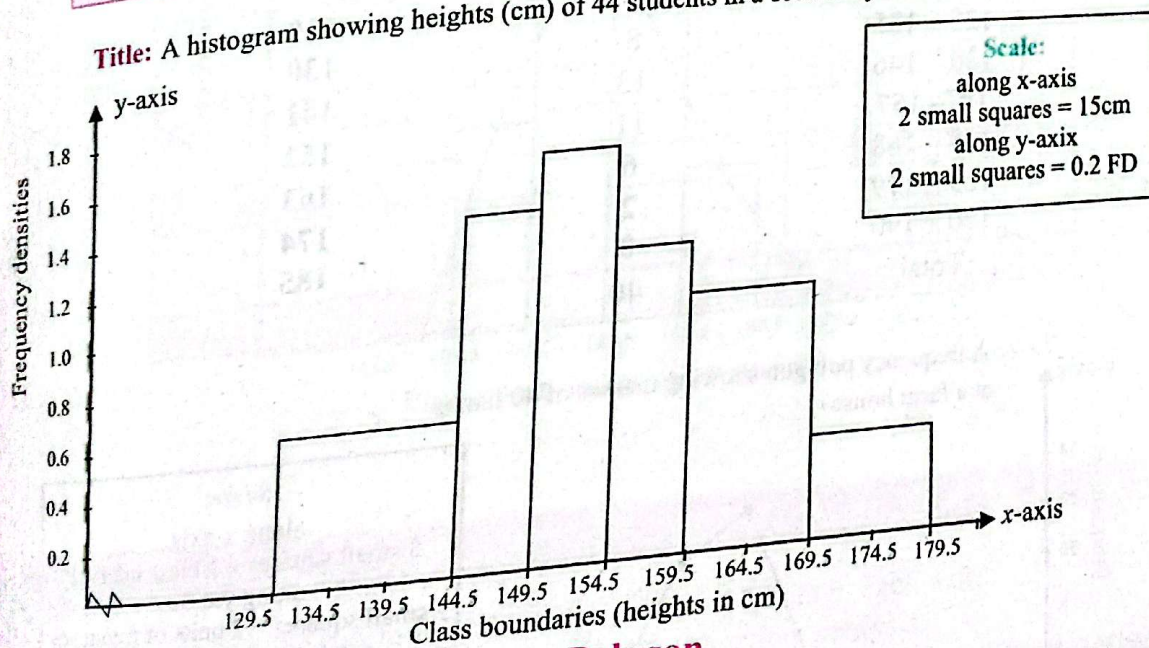
Height	130-144	145-149	150-154	155-159	160-169	170-179
No. of students	9	7	8	6	10	4

Represent the data by a histogram.

Solution:

Height (cm)	f	Size of class interval	F.D	Class boundaries
130 - 144	9	15	$9 \div 15 = 0.6$	129.5 - 144.5
145 - 149	7	5	$7 \div 5 = 1.4$	144.5 - 149.5
150 - 154	8	5	$8 \div 5 = 1.6$	149.5 - 154.5
155 - 159	6	5	$6 \div 5 = 1.2$	154.5 - 159.5
160 - 169	10	10	$10 \div 10 = 1$	159.5 - 169.5
170 - 179	4	10	$4 \div 10 = 0.4$	169.5 - 179.5
Total	44	—	—	—

Title: A histogram showing heights (cm) of 44 students in a secondary school.



11.1.3 Construction of a Frequency Polygon

Polygon is a many sided closed figure. To draw a frequency polygon mid points (class marks) are marked on the horizontal axis and frequencies are on the vertical axis. Points are plotted with class frequencies and their corresponding class marks. These points are joined by straight line segments. To complete a closed polygon we add extra classes at both ends with zero frequencies. In this way two extra points on x-axis are obtained on both ends. The points are joined by their nearer plotted points. Finally a closed frequency polygon is obtained. It is easy to draw a frequency polygon on a histogram. To do so we find mid points of the top edges of rectangles and join them by segments. Then find two extra points on both ends on x-axis to complete the closed polygon.

National Book Foundation

Example 4:

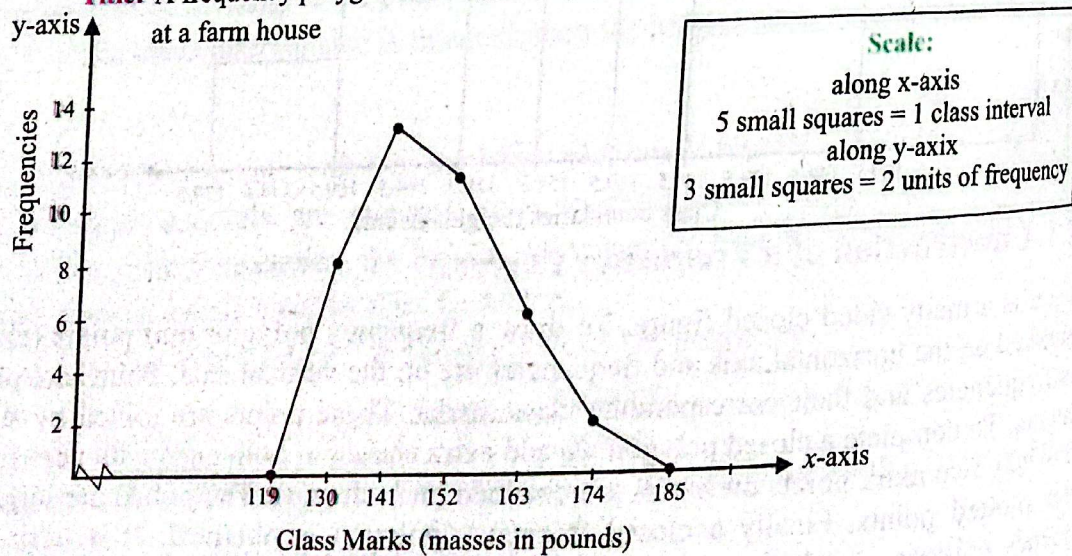
At a farm house there are 40 lambs, their masses are recorded as given below. Represent the following frequency distribution by a frequency polygon.

Masses in pounds	No. of lambs
125 – 135	8
136 – 146	13
147 – 157	11
158 – 168	6
169 – 179	2
Total	40

Solution:

Masses	Frequency	Class Marks
114 – 124	0	119
125 – 135	8	130
136 – 146	13	141
147 – 157	11	152
158 – 168	6	163
169 – 179	2	174
180 – 190	0	185
Total	40	

Title: A frequency polygon showing masses of 40 lambs at a farm house



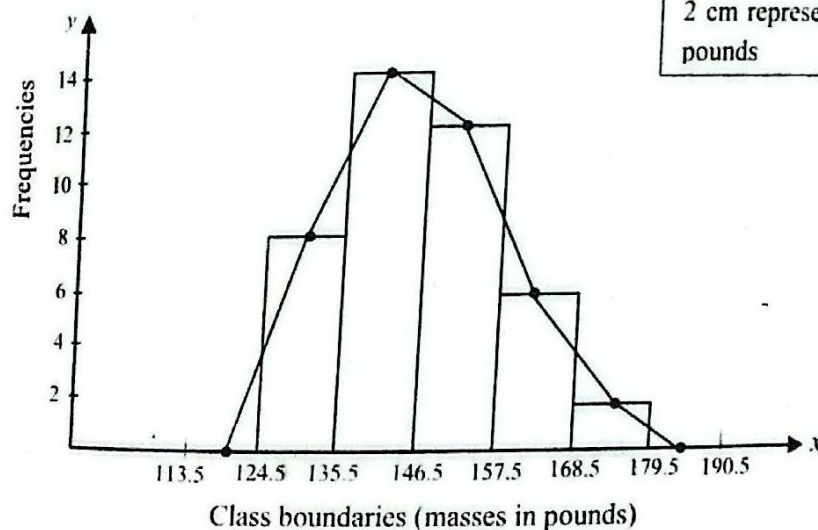
Example 5:
Use the frequency distribution in example 4 and draw a frequency polygon on a histogram.

Solution:

Masses	Frequency	Class boundaries	Class Marks
125 – 135	8	124.5 – 135.5	130
136 – 146	14	135.5 – 146.5	141
147 – 157	12	146.5 – 157.5	152
158 – 168	6	157.5 – 168.5	163
169 – 179	2	168.5 – 179.5	174

Title: A frequency polygon on a histogram showing masses of 40 lambs in pounds

Scale:
On horizontal axis
2 cm represents 11
pounds



Work Point

Read the following passage carefully and count number of letters in each word.

Statistics is a mathematical science pertaining to the collection, analysis, interpretation or explanation and presentation of data. It is applicable to a wide variety of academic disciplines from the physical and social sciences to the humanities".

Make a discrete frequency distribution for
(i) the number of letters in each word, (ii) the number of vowels in each word.

EXERCISE 11.1

1. The given table shows number of regular readers of a library who have completely read the books.

Class Interval (books)	Frequency (No. of readers)
1 – 10	5
11 – 20	4
21 – 30	8
31 – 40	9
41 – 50	2
51 – 60	2

- (i) What is the total number of readers in the data?
- (ii) Which group contains highest number of readers?
- (iii) Which group contains least number of readers?
- (iv) What is the lower limit of the last class?
- (v) What is the lower boundary of the last class?
- (vi) What is the size of the class interval?
- (vii) Find class marks (mid points) of all groups.

2. Number of gratitude cards made by 80 student of class 9 is given below.

79, 60, 74, 59, 55, 98, 61, 67, 89, 71, 71, 46, 63, 66, 69, 42, 75, 62, 71, 77, 78, 65, 87, 57, 78, 91, 82, 73, 65, 94, 48, 87, 62, 81, 63, 66, 65, 49, 45, 51, 69, 56, 84, 93, 63, 60, 68, 51, 73, 54, 50, 88, 76, 93, 48, 70, 40, 76, 95, 57, 63, 94, 82, 54, 89, 64, 77, 94, 72, 69, 51, 56, 67, 88, 81, 70, 81, 54, 66, 87

- (i) Prepare a frequency table to represent the above data using the classes 40 – 49 etc.
- (ii) Add cumulative frequency column in the table.
- (iii) How many students made less than 70 cards?
- (iv) What percent of students made less than 50 cards?
- (v) Which group has the greatest frequency?
- (vi) What is the size of the class interval?

3. The number of medals won by 45 players in a certain sports festival is given as :
0, 2, 1, 0, 1, 2, 3, 5, 6, 3, 2, 1, 3, 4, 2, 6, 1, 5, 2, 4, 3, 0, 1, 2, 3, 0, 0, 2, 3, 4, 1, 5, 6, 2, 4, 5, 1, 3, 4, 6, 2, 3, 1, 2, 5.

Prepare a discrete frequency distribution. Also make a column of cumulative frequencies.

4. The ages of workers in a factory were recorded as below.

Ages in Years	20 – 24	25 – 29	30 – 34	35 – 39	40 – 44	45 – 49
No. of workers	5	16	12	10	8	4

Draw a frequency polygon to represent the data.

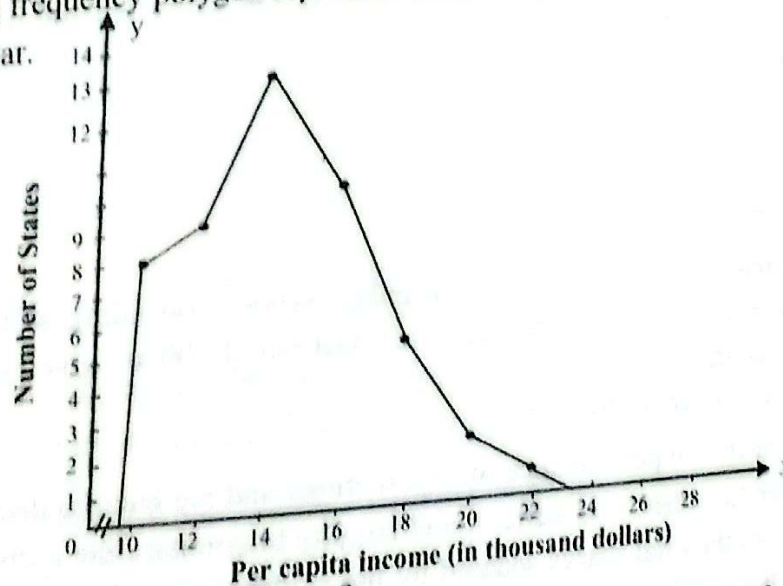
5. Represent the following data by a histogram.

Class Interval	0 - 5	5 - 15	15 - 30	30 - 40	40 - 45
Frequency	5	12	16	10	2

6. In a saving group, there are 400 members, and the number of savings certificates held by them is shown in the following table. Construct a histogram of the distribution of saving certificates.

Class Interval	1 - 50	51 - 100	101 - 150	151 - 200	201 - 300	301 - 400	401 - 500
No. of members	10	15	30	40	120	100	85

7. The following frequency polygon represents the per-capita income of some states, recorded in a particular year.



- What is the total number of states?
 - What is the number of states which have the least per-capita income?
 - What is the number of states having highest per-capita income?
 - What percent of states have per-capita income \$14000 and \$16000?
 - How many states have per-capita income \$18000 and above?
 - Represent the frequency polygon by a frequency table.
8. The masses measured to the nearest kg of 50 boys are given in the frequency distribution as :

Mass (kg)	60 - 64	65 - 69	70 - 79	80 - 89	90 - 94	95 - 99
f	2	6	12	14	10	6

- Construct a histogram.
 - Construct a frequency polygon.
9. At a sale, number of items placed according to the price range is given below:

Price(Rs)	600 - 999	1000 - 1999	2000 - 2499	2500 - 2999	3000 - 3499
f	50	70	75	65	70

Construct a histogram.



11.2 Measures of Central Tendency

In this unit we are looking for a better representation of data. The purpose of summarizing a data is to draw meaningful results, to analyze the data and to compare two or more data having same characteristics. When the data have been arranged into a frequency distribution the information contained in it is easily understood. The data becomes clearer when it is represented by graphs. Still we need a simpler way to represent a data in more efficient way to analyze and understand better. A single value that represents a set of data is called an average. Since the averages tend to lie in the center of a data, they are called measures of central tendency. They are also called measures of location because they tend to be located at the center of the data. Here we will discuss 3 very common measures of central tendency.

- (i) Arithmetic Mean
- (ii) Median
- (iii) Mode

11.2.1 (a) Arithmetic Mean

There are two sections A and B of class 9 in a public school consisting of 10 and 12 students respectively. The marks secured by each student in a test out of 100 are given below :

A : 35, 40, 90, 45, 50, 80, 70, 65, 55, 50

B : 25, 50, 30, 35, 60, 45, 45, 85, 90, 65, 45, 35

It is required to compare the performance of two sections and we have to decide that which one of them is better than the other. It is difficult and lengthy to compare the individual performance of students. Let us see the total marks secured by the students in each section. The total of both sections are same. It is not good to say that performance of both sections is same, because they have different number of students. We have to compare sum of scores and numbers of students in each section simultaneously. To do this we will find the ratio of sum of scores and number of students in each section separately and then we will compare the two results obtained.

In section A, sum of scores: number of students = $610:10 = 61$

In section B, sum of scores: number of students = $610:12 = 50.83$

As $61 > 50.83$ so average performance of section A is better than that of section B.

“The ratio of sum of values of a data and number of values is called an arithmetic mean.”

Arithmetic Mean for Ungroup Data

The values of data are usually denoted by x . In case of more than one data, their values are denoted by x , y or z etc. The sum of the values of the data ' x ' is denoted by " $\sum x$ " and number of values is denoted by n . The arithmetic mean of data x is symbolized a $\bar{x}$ and by above

definition:

$$\bar{x} = \frac{\sum x}{n} = \frac{\text{sum of values}}{\text{numbers of values}}$$

If values of data occur more than once, the sum of values can be found by using frequencies e.g.
 if $x = 2, 3, 1, 2, 2, 3, 2, 3, 4, 2, 5, 2, 3, 4, 5, 2, 1, 3, 5, 1$
 Then, $\sum x = 2+3+1+2+2+3+2+3+4+2+5+2+3+4+5+2+1+3+5+1 = 55$
 In this data frequency of 1 is 3 i.e. 1 occurs three times. Similarly frequencies of 2, 3, 4, 5, are 7, 5, 2, 3 respectively.

$\sum x = 1 \times 3 + 2 \times 7 + 3 \times 5 + 4 \times 2 + 5 \times 3 = 3 + 14 + 15 + 8 + 15 = 55$
 Also $n = \text{number of values} = \text{sum of frequencies} = 3 + 7 + 5 + 2 + 3 = 20$
 So, $\bar{x} = \frac{\sum x}{n} = \frac{55}{20} = 2.75$

The above data can be summarized in frequency distribution as follows:

x	1	2	3	4	5	Total
f	3	7	5	2	3	$\sum f = 20$
fx	$1 \times 3 = 3$	$2 \times 7 = 14$	$3 \times 5 = 15$	$4 \times 2 = 8$	$5 \times 3 = 15$	$\sum fx = 55$

The arithmetic mean is $\bar{x} = \frac{\text{sum of values}}{\text{numbers of values}} = \frac{\sum fx}{\sum f} = \frac{55}{20} = 2.75$

The arithmetic mean for a grouped data can also be found by using class marks ' x ' and class frequencies ' f ' with the help of formula $\bar{x} = \frac{\sum fx}{\sum f}$, if size of class interval is greater than 1

Example 6: Find the arithmetic mean of the following data

Marks	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No. of students	2	5	7	3	8

Solution:

Marks	f	Class marks	fx
0 - 10	2	5	10
10 - 20	5	15	75
20 - 30	7	25	175
30 - 40	3	35	105
40 - 50	8	45	360
Total	$\sum f = 25$	-	$\sum fx = 725$

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{725}{25} = 29$$

11.2.1 (b) Median

When data is arranged in some order, it can be represented by a single value that lies exactly at the centre of data. "The central or middle most value of an arranged data is called its median represented by $\tilde{x}$ read as x tilde .

Median for Ungroup Data

- If the number of values of a data is odd, then the middle value of the arranged data is the median.
 - If the number of values of a data is even, then there are two middle values, the arithmetic mean of these two middle values is called median of the data. These definitions of median can be expressed for an ungrouped data as following formulas.
- (i) If n (number of values) is odd, then median = $\frac{n+1}{2}$ th value of the data
- (ii) If n is even, then Median = Arithmetic mean of the two middle values
- $$= \frac{1}{2} \left\{ \frac{n}{2} \text{ th value} + \frac{n+2}{2} \text{ th value} \right\}$$

Example 7:

The selling prices of the seven articles sold by a shopkeeper in rupees were 1075, 1050, 2050, 1030, 2045, 1090 and 1530. Find the median selling price of these articles.

Solution:

Arranging these values in the ascending order as 1030, 1050, 1075, 1090, 1530, 2045, 2050.
Here $n = 7$ (odd), so

$$\text{Median} = \frac{n+1}{2} \text{ th value} = \frac{7+1}{2} \text{ th value} = \frac{8}{2} \text{ th value} = 4^{\text{th}} \text{ value} = 1090$$

So, median selling price of seven articles is Rs.1090

Example 8:

Find the median of the following set of values.

40, 35, 45, 60, 50, 35, 40, 45, 35, 50

Solution: Arrange the numbers in ascending order as 35, 35, 35, 40, 40, 45, 45, 50, 50, 60

Here $n = 10$ (even), so

Median = $\tilde{x}$ = half of sum of two middle most values

$$\begin{aligned} &= \frac{1}{2} \left\{ \frac{n}{2} \text{ th value} + \frac{n+2}{2} \text{ th value} \right\} \\ &= \frac{1}{2} \left\{ \frac{10}{2} \text{ th value} + \frac{10+2}{2} \text{ th value} \right\} \\ &= \frac{1}{2} (5^{\text{th}} \text{ value} + 6^{\text{th}} \text{ value}) \\ &= \frac{1}{2} (40 + 45) = \frac{85}{2} = 42.5 \end{aligned}$$

Median for Grouped Data
 In a grouped data or frequency distribution the median is taken as $\frac{n}{2}$ th value. To find median for a grouped data we first find the group which contains the median. To locate the median group we form cumulative frequency table. The formula used to find median of a group data is

$$\text{Median} = \tilde{x} = l + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

Where l = lower class boundary of median group
 h = size of class interval

f = frequency of median group
 $n = \Sigma f$ = Number of items

c = Cumulative frequency of class preceding the median class.

Example 9: Marks obtained by 100 students in an examination are given below,

Marks	10 - 19	20 - 29	30 - 39	40 - 49	50 - 59
No of students	5	25	40	20	10

Find the median marks.

Solution: Median = $\frac{n}{2}$ th value = $\frac{100}{2}$ th value = 50th value.

Marks	f	Class Boundaries	c.f
10 - 19	5	9.5 - 19.5	5
20 - 29	25	19.5 - 29.5	c = 30
30 - 39	$f = 40$	29.5 - 39.5	70
40 - 49	20	39.5 - 49.5	90
50 - 59	10	49.5 - 59.5	100

→ median group

From the above table, it is clear that 50th value lies in the group 29.5 - 39.5.
 So 29.5 - 39.5 is the median group.

Median = $l + \frac{h}{f} \left(\frac{n}{2} - c \right)$ where $l = 29.5$, $h = 10$, $n = 100$, $f = 40$, $c = 30$

$$\begin{aligned} \text{Median} &= 29.5 + \frac{10}{40} \left(\frac{100}{2} - 30 \right) \\ &= 29.5 + \frac{1}{4} (50 - 30) \\ &= 29.5 + \frac{20}{4} \\ &= 29.5 + 5 = 34.5 \text{ marks} \end{aligned}$$

Danger zone !!!
 Most of the students take c from the median group which gives the negative result .
 remember c is taken from the group which comes before median group

11.2.1 (c) Mode

The number of pairs of shoes sold in a day according to their sizes at a shoe store are given below.

Sizes	6	7	8	9	10	11
No. of pairs sold	14	25	45	50	30	16

According to the data, the most commonly used shoes are of size 9. In other words we can say that the average size of shoes sold in a day at a shoe store is 9. The average found in this way is called mode. "The most frequent (repeated) value of a data is called mode of the data." If two or more values occur the same numbers of times but more frequently than any of the other values, then there is more than one mode. In this respect, mode differs from mean and median because they have unique values. If each value of data occurs same numbers of times, then there is no mode.

Example 10: Find mode of following data sets.

(i) 30, 25, 40, 30, 45, 25, 30, 40, 30

(ii) 1, 2, 3, 2, 1, 3, 1, 2

(iii) 125, 130, 115, 125, 135, 120, 135, 130, 120

Solution:

(i) As 30 is the most frequent value so it is mode of the data. The data is called unimodal.

(ii) As the values 1 & 2 occur equal number of times, so there are 2 modes. This data is called bimodal data.

(iii) As the values 120, 125 and 130 occur two times each and frequencies of remaining values are lesser. So 120, 125 and 130 are three modes of the data. This data is called trimodal data.

Mode for Group Data

To find mode for a frequency distribution or a group data, we first find the modal class. A class or group having largest number of items (frequency) is called modal group and it contains mode of the data. The mode of a grouped data can be found by the formula.

$$\text{Mode} = l + \frac{(f_m - f_1)}{2f_m - f_1 - f_2} \times h$$

Where l = lower class boundary of the modal class.
 f_m = frequency of the modal class (the highest frequency)
 f_1 = frequency of the class preceding the modal class
 f_2 = frequency of the class following the modal class
 h = size of class interval

Note: If first group of the distribution have the largest frequency, then $f_1 = 0$ and if modal group is the last group of frequency distribution, then $f_2 = 0$. But these are very rare cases.

Example 11:
Fifty workers of a factory are distributed in 7 groups according to their wages in rupees per hour.

Wages in Rupees	25-29	30-34	35-39	40-44	45-49	50-54	55-59
No of workers	2	4	8	20	6	6	4

Find mode of the distribution.
Solution:

Wages	f	Class boundaries
25 - 29	2	24.5 - 29.5
30 - 34	4	29.5 - 34.5
35 - 39	8	34.5 - 39.5
40 - 44	20	39.5 - 44.5 → modal class
45 - 49	6	44.5 - 49.5
50 - 54	6	49.5 - 54.5
55 - 59	4	54.5 - 59.5

As 20 is the largest frequency, so (39.5 - 44.5) is the modal group.
 $l = 39.5, f_m = 20, f_1 = 8, f_2 = 6, h = 5$

$$\text{Mode} = l + \frac{(f_m - f_1)}{2f_m - f_1 - f_2} \times h$$

$$\text{Mode} = 39.5 + \frac{(20 - 8)}{2 \times 20 - 8 - 6} \times 5$$

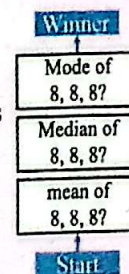
$$= 39.5 + \frac{12}{40 - 14} \times 5$$

$$= 39.5 + \frac{12}{26} \times 5 = 39.5 + \frac{60}{26}$$

$$= 39.5 + 2.31 = 41.81 \text{ Rupees}$$

Math Play Ground

1. Make such hopscotch in the play ground.
2. Ask a player to start hoping, doing calculations in each box.
3. Decide the winner if he completes without falling.



Example 12:

Find median and mode for the following discrete frequency distribution.

x	1	2	3	4	5
f	3	4	5	2	2

Solution:

Here $n = \sum f = 16$

Median = class containing $\frac{n}{2}$ th observation = $\frac{16}{2}$ th observation = 8th observation

x	f	c.f.
1	3	3
2	4	7
3	5	12
4	2	14
5	2	16

8th observation

So Median = 3 (8th observation lies in the group where c.f. is 12.)

Hence Mode = most frequent observation = 3

11.2.1 (d) Empirical Relation between Mean, Median and Mode

If any two of mean, median and mode of a data are given, then we can find an approximate value of the third measure of central tendency by using an empirical relation.

$$\text{mode} = 3 \text{ median} - 2 \text{ mean}$$

$$\text{Or } \text{median} = \frac{1}{3} (\text{mode} + 2 \text{ mean})$$

$$\text{Or } \text{mean} = \frac{1}{2} (3 \text{ median} - \text{mode})$$

11.2.2 Weighted Mean

If it is required to find the average of some values which are not equally important, then we assign them certain numerical values to express their relative importance. These numerical values are called weights. To find arithmetic mean of data whose values are not equally important, we use the weights instead of frequencies.

If x denotes the items of a data and w denotes their respective weights, then weighted arithmetic mean is formulated as: $\bar{x}_w = \frac{\sum wx}{\sum w}$

Example 14: The following are the monthly expenditure of a house according to their weights.

Items	Expenditure (Rs)	Weight relative importance
Food	8,000	20
House rent	5,000	10
Clothing	1,000	6
Utility bills	900	4
Education	600	2
Miscellanies	100	1

Find the weighted mean.

Solution:

Items	$x = \text{Expenditure (Rs)}$	$w = \text{Weight}$	wx
Food	8,000	20	160000
House rent	5,000	10	50000
Clothing	1,000	6	6000
Utility bills	900	4	3600
Education	600	2	1200
Miscellanies	100	1	100
Total	—	43	220900

$$\text{Weighted mean: } \bar{x}_w = \frac{\sum wx}{\sum w} = \frac{220900}{43} = \text{Rs.}5137.21$$

EXERCISE 11.2

Find arithmetic mean for the following data by definition

- (i) $x = 2, 4, 6, 8, 10, 12$
 (ii) $y = 3, 4, -1, 7, -8, -5, 0$
 (iii) $z = 0, 4, 8, 12, 16, 20, 24, 28$
 (iv) $u = 3.1, 4.2, 5.3, 6.4, 7.5, 8.6, 9.7, 10.8$
 (v) $v = 5, 5, 5, 5, 5, 5, 5, 5$

Following are the scores made by two batsmen A and B in a series of innings:

A	12	15	6	73	7	19	199	36	84	29
B	47	12	76	48	4	51	37	48	13	0

Find arithmetic mean of scores of both players and state who is better as run getter?

Marks of four students of class 9 in four subjects out of 100 are given below. The weights of subjects according to their importance are also given.

Subjects	Weights	Marks			
		Hanzala	Musa	Zaki	Saleh
Mathematics	4	70	80	90	95
Physics	3	95	60	55	70
Chemistry	2	80	65	50	60
Biology	1	55	45	65	60

Find weighted mean of marks of each student.

Following frequency distribution represents number of liters of fuel consumed against the distance in km. Find arithmetic mean of this data set.

Class intervals	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
Frequencies	5	7	10	6	2

Find arithmetic mean from the following data of number of a detergent packs sold against its mass in kg.

Classes(kg)	1 – 9	10 – 18	19 – 27	28 – 36	37 – 45
No. of packs	10	15	18	15	12

Find median for the following data.

- (i) 1, 4, 2, 5, 3, 7, 6 (ii) $\pm 2, \pm 4, \pm 6, \pm 8$ (iii) 0, 1, -2, -3, 4, 5, 6, 3
 (iv) 4, 3, 1, -3, 2, -3, 3, 4, 1 (v) 4, 4, 4, 4, 4, 4

11.2.1 (d) Empirical Relation between Mean, Median and Mode

If any two of mean, median and mode of a data are given, then we can find an approximate value of the third measure of central tendency by using an empirical relation.
$$\text{mode} = 3 \text{ median} - 2 \text{ mean}$$

Or
$$\text{median} = \frac{1}{3} (\text{mode} + 2 \text{ mean})$$

Or
$$\text{mean} = \frac{1}{2} (3 \text{ median} - \text{mode})$$

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If it is required to find the average of some values which are not equally important, then we assign them certain numerical values to express their relative importance. These numerical values are called weights. To find arithmetic mean of data whose values are not equally important, we use the weights instead of frequencies.

If x denotes the items of a data and w denotes their respective weights, then weighted arithmetic mean is formulated as:
$$\bar{x}_w = \frac{\sum wx}{\sum w}$$

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Clothing	1,000	6
Utility bills	900	4
Education	600	2
Miscellanies	100	1

Find the weighted mean.

Solution:

Items	x = Expenditure (Rs)	w = Weight	wx
Food	8,000	20	160000
House rent	5,000	10	50000
Clothing	1,000	6	6000
Utility bills	900	4	3600
Education	600	2	1200
Miscellanies	100	1	100
Total	—	43	220900

$$\text{Weighted mean: } \bar{x}_w = \frac{\sum wx}{\sum w} = \frac{220900}{43} = \text{Rs.}5137.21$$

EXERCISE 11.2

Find arithmetic mean for the following data by definition

- (i) $x = 2, 4, 6, 8, 10, 12$
 (ii) $y = 3, 4, -1, 7, -8, -5, 0$
 (iii) $z = 0, 4, 8, 12, 16, 20, 24, 28$
 (iv) $u = 3.1, 4.2, 5.3, 6.4, 7.5, 8.6, 9.7, 10.8$
 (v) $v = 5, 5, 5, 5, 5, 5, 5, 5$

Following are the scores made by two batsmen A and B in a series of innings:

A	12	15	6	73	7	19	199	36	84	29
B	47	12	76	48	4	51	37	48	13	0

Find arithmetic mean of scores of both players and state who is better as run getter?

Marks of four students of class 9 in four subjects out of 100 are given below. The weights of subjects according to their importance are also given.

Subjects	Weights	Marks			
		Hanzala	Musa	Zaki	Saleh
Mathematics	4	70	80	90	95
Physics	3	95	60	55	70
Chemistry	2	80	65	50	60
Biology	1	55	45	65	60

Find weighted mean of marks of each student.

Following frequency distribution represents number of liters of fuel consumed against the distance in km. Find arithmetic mean of this data set.

Class intervals	0 - 5	5 - 10	10 - 15	15 - 20	20 - 25
Frequencies	5	7	10	6	2

Find arithmetic mean from the following data of number of a detergent packs sold against its mass in kg.

Classes(kg)	1 - 9	10 - 18	19 - 27	28 - 36	37 - 45
No. of packs	10	15	18	15	12

Find median for the following data.

- (i) 1, 4, 2, 5, 3, 7, 6 (ii) $\pm 2, \pm 4, \pm 6, \pm 8$ (iii) 0, 1, -2, -3, 4, 5, 6, 3
 (iv) 4, 3, 1, -3, 2, -3, 3, 4, 1 (v) 4, 4, 4, 4, 4, 4

7. In a practical of geometry and shapes students are required to construct a circle of any radius between 0cm and 2cm. the data given below is recorded.

Radius (xcm)	1.8	1.6	1.4	1.2	1.0	0.8	0.6	0.4	0.2
No. of circles (f)	6	9	15	24	33	20	13	7	5

Find median radius of the circle.

8. At an old age house, there are people of different age groups as given below.

Classes intervals	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
f	3	8	11	5	4

Find an average median age of people residing in that old age house.

9. Find mode for the following data.

(i) 1, 2, 3, 4, 5, 6

(ii) 2, 4, 2, 3, 2, 5, 3, 2, 5, 4, 2

(iii) 120, 130, 140, 225, 125, 225, 120

(iv) 2, 4, 3, 5, 5, 3, 4, 2

10. Find mean, median and mode for the following discrete frequency distribution.

x	0	1	2	3	4	5
f	3	5	9	4	2	1

11. Find modal value for the following frequency distributions.

(a)

Classes	1 – 4	5 – 8	9 – 12	13 – 16	17 – 20
f	9	8	10	6	7

(b)

Classes	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35
Frequency	7	6	3	2	1

11.3 Probability

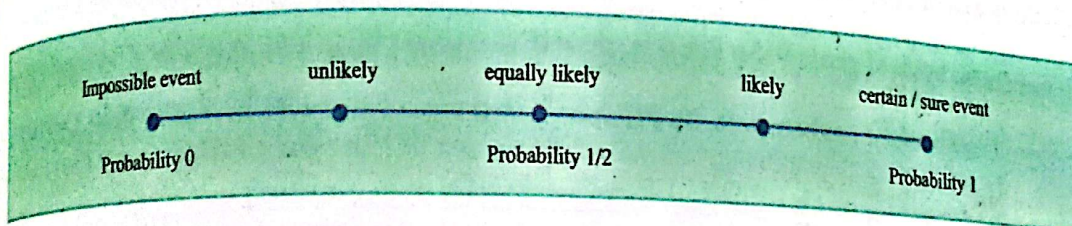
We often come across such situations where we have to make guess about occurrence or non occurrence of something that it would happen or not. For instance we often make such statements :

- 40% students of class 9 are predicted to be absent tomorrow because of severe heat wave.
- You will probably win Maths Olympiad this year .
- I hope you will get at least 90% marks in your board exam .

In these statements one can notice the uncertainty or doubt of occurrence. These are few of the guesses made in our daily life for the possibility of an event that the event is likely to happen or not. Probability is synonymous to possibility. Simply speaking the measure of chance of an event is called probability and its values are from 0 to 1. For example

- Probability of a new prophet after Hazrat Muhammad (SAW) is 0.
- Probability that Haleema will win the toss among Haleema, Hajra & Haani is $1/3$.
- Probability of sun setting in west is 1.

In these statements we have seen that probability is between 0 and 1 both inclusive. This range can also be presented pictorially for better understanding as



In previous classes you have learnt about various terms necessary to understand the problems of probability i.e.

Experiment, trial, outcome, sample space, event, impossible event, sure event, equally likely, likely, unlikely, mutually exclusive events, exhaustive events, independent and dependent events etc. Here we will discuss single event and events not happening in a bit detail.

11.3.1 Single Event Probability

An event which has single sample point is called a single event or simple event or singular event. Eg in tossing a coin the sample space is $\{H, T\}$, where $E_1 = \{H\}$ and $E_2 = \{T\}$, as only one out of 2 outcomes is possible. Here both E_1 and E_2 are single events.

Example 15: Write sample space for single roll of a die and enlist all the events. Discuss whether any compound event is present in this case.

Solution: Here we notice if a single die is rolled once then any one of the outcome may occur from 1, 2, 3, 4, 5 and 6. Therefore $S = \{1, 2, 3, 4, 5, 6\}$

Now all possible events are $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$

from events listed above, its evident that all events are single. So no compound event in this case.

The probability of such events is obtained by the formula:

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes.}}$$

Example 16: Write sample space for single spin of a spinner of 3 primary colours and enlist all the events. Discuss probabilities of all events.

Solution: Here $S = \{\text{blue, red, green}\}$, the pointer of the spinner can stop at only one of the colour so $E_1 = \{\text{blue}\}$, $E_2 = \{\text{red}\}$, $E_3 = \{\text{green}\}$

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Example 17: Ali picked a book at random from a library shelf which had 3 history books of grade 6, 4 Geography books grade 7, 9 Science books of grade 8, 3 science books of grade 7, 6 Geography books of grade 8 and 5 science books of grade 6. Find the probability of getting:

- a Science book
- a Geography book
- a book of grade 7
- a book of grade 9
- a book of Mathematics

Solution: Here total number of books is 30. Let $E_1, E_2, \dots, E_5$ be the events of picking science books, Geography books, grade 7 books, grade 9 books and Mathematics books respectively.

- total science books are 17. So $P(E_1) = \frac{17}{30}$
- total geography books are 10. So $P(E_2) = \frac{10}{30} = \frac{1}{3}$
- total books of grade 7 are 7. So $P(E_3) = \frac{7}{30}$
- there is no book of grade 9. So $P(E_4) = \frac{0}{30} = 0$
- there is no book of mathematics. So $P(E_5) = \frac{0}{30} = 0$

11.3.2 Probability of Complementary Events

Two events which can not occur simultaneously are called complementary events.

If 4 boys Affan, Jaffar, Hassan and Maaz are playing ONO, then probability of winning of each boy is $\frac{1}{4}$. If the event of winning of jaffar is represented by J , then $P(J) = \frac{1}{4}$. Now the event that Jaffar won't win is represented by J' and $P(J') = 1 - P(J) = 1 - \frac{1}{4} = \frac{3}{4}$

The two events in which, one shows non occurrence of other are called complementary events.

REMEMBER: Sum of Probabilities of 2 complementary events is unity.

Example 18: There are 12 teachers in maths department among them 2 are higher secondary school teachers, 4 are secondary school teachers and 6 are senior elementary school teachers. If good performance award is to be given to only one teacher then find probability

- That a secondary school teacher will get the award
- That a secondary school teacher will not get the award

Solution: Here total number of teachers is 12. If this is assumed that all teachers are equally hardworking, dedicated and showing excellent performance throughout their career then the events will be equally likely. Let E and E' be the events that a secondary teacher will get the award and will not get the award respectively.

- Probability that a secondary teacher will get the award is $P(E) = \frac{4}{12} = \frac{1}{3}$
- Probability that a secondary teacher will not get the award is $P(E') = 1 - P(E) = 1 - \frac{1}{3} = \frac{2}{3}$

When theoretical probability is not certain, we use relative frequency to estimate probabilities, as in most of the cases it is hard to predict the probability of an event happening finely.

Relative frequency is defined as the ratio between number of times an event is occurring in an experimental trial and total number of trials in the experiment. i.e.

- Relative Frequency =
$$\frac{\text{Frequency of an event occurring}}{\text{Total number of trials in an experiment}}$$
- The relative frequency is often calculated after performing experiment so it is often called experimental probability.
 - More the number of experiments done, closer the relative frequency becomes to the theoretical probability.
 - Empirical probability is another name of relative frequency.

Example 19: A spinner of 5 equally likely colours red, blue, green, yellow and white is spun 50 times and it landed 10 times on red, 12 times on blue, 9 times on green, 9 times on yellow and 10 times on white colour. Find the relative frequency for the event that it lands on green colour.

Solution: Total number of trials = $n = 50$
Number of favourable outcomes for landing on green colour = $f = 9$

$$\text{Relative Frequency} = \frac{\text{Frequency of an event occurring}}{\text{Total number of trials in an experiment}} = \frac{f}{n} = \frac{9}{50}$$

Example 20: A fair coin is tossed 150 times and head comes up 70 times. What is the relative frequency of getting head up.

Solution: Total number of trials = $n = 150$ number of favourable outcomes for getting head up = $f = 70$

$$\text{Relative Frequency} = \frac{\text{Frequency of an event occurring}}{\text{Total number of trials in an experiment}} = \frac{f}{n} = \frac{70}{150} = 0.466\ldots$$

11.3.4 Expected Frequency

In probability we often come across the situation where the probability of an event becomes slightly different from its theoretical probability. The expected frequency is defined as how often (we think) an event should happen on the basis of theoretical probability or on the basis of previously happened experiments (relative frequency).

For instance, If Ali rolled a fair die it is expected that 4 will happen $1/6$ times. If he rolls it 120 times, then he expects 4s to happen $120 \times 1/6 = 20$ times. Hence expected frequency is given by the formula: Expected Frequency = Number of trials $\times$ probability of the event = $n \times p$

Example 21: 5 students out of a section of 30 are expected absent everyday in summer camp. What number of students is expected absent every day from 12 such sections of the same strength.

Solution: Total number of students in 12 sections = $n = 12 \times 30 = 360$

$$\text{Probability of absent students} = p = \frac{5}{30} = \frac{1}{6}$$

$$\text{Expected frequency} = \text{Number of trials} \times \text{probability of the event} = n \times p = 360 \times \frac{1}{6} = 60$$

Example 22: If Probability of a person to be Hafiz e Quran in a village is $\frac{3}{10}$. What do you expect the number of Hafiz e Quran in 2500 such villages?

Solution: Total number of villages = $n = 2500$ and Probability = $p = \frac{3}{10}$
 Expected No. of Hafiz e Quran = number of villages $\times$ probability of the event
 $= n \times p = 2500 \times \frac{3}{10} = 750$

EXERCISE 11.3

- A letter is chosen randomly from the word 'ALLAH'. Now find the probability of getting a/an
 (i) vowel (ii) H (iii) L (iv) consonant
- A garment factory shipped an order which contained 7000 jackets, 2000 sweaters, and 3000 trousers in which 7 trousers, 5 jackets and 9 sweaters are faulty. If the quality inspector unpacked one item at random, find the probability that it is
 (i) a trouser (ii) not a jacket (iii) a faulty item
- A number wheel is divided into 8 equal sectors labelled as pansy, lily, orchid, tulip, jasmine, rose, marigold and sunflower. Maazz spins the wheel once. Find the probability that pointer:
 (i) Stops at rose (ii) stops at a 4 letter flower name
 (iii) Doesnot stop at marigold (iv) stops at a 3 letter flower name
 (vi) stops at a flower name
- A decagonal die labelled 4, 4, 4, 4, 5, 5, 6, 7, 8, 8 is rolled once. Find the probability of getting:
 (i) 4 (ii) even number (iii) multiple of 4
 (iv) not a 7 (v) odd number (vi) prime number
 (vii) LCM of 4 and 8 (viii) HCF of 4 and 8 (ix) factor of 12
- A one digit whole number is chosen at random. Find the probability that it is:
 (i) less than 5 (ii) greater than 10
 (iii) not the largest 1 digit number (iv) additive identity of whole numbers
 (v) HCF of 3 and 5 (vi) multiplicative identity of real numbers
 (vii) not a prime number (viii) factor of 9
- A coin is tossed 10 times. Complete the given table and answer the questions below the table.

Event	frequency	Relative frequency
head	6	
tail	4	

- What is the expected frequency of getting head, if it was tossed 520 times?
- What is the expected frequency of getting tail, if it was tossed 305 times?

7. Complete the table and give answers, if a die is rolled 120 times

Number	Frequency	Relative frequency
1	15	...
2	25	...
3	15	...
4	20	...
5	30	...
6	15	...

- if it was rolled 6000 times, how many times occurrence of a 6 is expected?
- if it was rolled 3000 times, how many times occurrence of a 1 is expected?
- if it was rolled 400 times, how many times occurrence of a 2 is expected?

8. There are 6 girls sections of class 9 in a public school, namely Teal, orchid, mauve, hazel, zaffre and denim. Only one girl is to be chosen randomly from all sections for national science olympiad.
- Find the probability of selecting a girl from hazel section.
 - Find the probability of not selecting a girl from hazel section.
 - Verify that answers of both parts a and b add up to unity.
 - If 6 girls are to be selected, find expected frequency of selecting from orchid section.
 - If 60 girls are to be selected, what number of girls is expected to be chosen from Teal.

KEY POINTS

- Statistics is science of collecting and analyzing numerical data.
- Facts and statistics collected for analysis is called data.
- A table consisting of values of a data along with their frequencies is called a frequency table.
- In a group data, the table consisting of class intervals and their respective frequencies is called frequency table or a frequency distribution.
- The representation of data by a chart having adjacent rectangular bars is called histogram.
- The representation of data by a closed polygon in which class marks are plotted along x-axis and frequencies along y-axis is called a frequency polygon.
- A single value which can represent the whole data is called measure of central tendency, measure of location or an average.
- Measures of central tendency discussed in this grade are arithmetic mean, median and mode.
- Arithmetic mean is a ratio between the sum of items and number of items of the data.
- The middle most value of an arranged data is called median average or simply median.
- The most frequent value of the data is called modal average or mode.
- To find arithmetic mean of data whose values are not equally important, we use the weights instead of frequencies. This is called weighted arithmetic mean.

- The measure of chance of an event is called probability.
- The values of probability are from 0 to 1.
- The two events in which one shows non occurrence of other are called complementary events.
- Relative frequency is defined as the ratio between no. of times an event is occurring in an experimental trial and total number of trials in the experiment.
- The expected frequency of an event is the frequency we expect to see based on probabilities.

MISCELLANEOUS EXERCISE II

1. Encircle the correct option in the following.

- Which of the following is a class mark of the interval (10 – 15)?
(a) 10 (b) 12.5 (c) 15 (d) 16
- What is size of class interval (4 – 7)?
(a) 4 (b) 5 (c) 6 (d) 7
- Which of the following is chart of adjacent rectangles?
(a) bar graph (b) Frequency polygon (c) histogram (d) ogive
- Which of the following is measure of central tendency?
(a) variance (b) standard deviation (c) range (d) arithmetic mean
- Which of the following is a formula of arithmetic mean for grouped data?
(a) $\frac{\sum x}{n}$ (b) $l + \frac{h}{f}(\frac{n}{2} - c)$ (c) $\frac{\sum fx}{\sum f}$ (d) $l + \frac{(f_m - f_1) \times h}{2f_m - f_1 - f_2}$
- If arithmetic mean of 25 values is 10, then what is sum of values?
(a) 250 (b) 125 (c) 25 (d) 2.5
- What is median of the data 4, 3, 0, 2, 1?
(a) 0 (b) 2 (c) 3 (d) 4
- Which measure of central tendency can have more than one value?
(a) weighted mean (b) median (c) mode (d) arithmetic mean
- The probability of getting M in 'MUHAMMAD' is
(a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{3}{5}$ (d) none of these
- Probability of picking an ace from a well shuffled pack of 52 playing cards is
(a) $\frac{1}{52}$ (b) $\frac{1}{13}$ (c) $\frac{4}{13}$ (d) none of these
- A normal fair die is rolled 6000 times. The expected number of 5 is
(a) 100 (b) 5000 (c) 1000 (d) none of these
- A fair coin is tossed 500 times. Expected number of tails is
(a) 100 (b) 250 (c) 1000 (d) none of these
- In a group of 5 people, 4 like peach juice. The expected no. of people in a population of 1200 who like peach juice is
(a) 240 (b) 600 (c) 960 (d) none of these

1. How many times Haani should toss a fair coin if he expects to get 100 tails
 (a) 100 (b) 200 (c) 1000 (d) none of these
2. An event which can never happen is called
 (a) sure event (b) certain event (c) possible event (d) impossible event
3. Two such events whose probabilities are $\frac{1}{2}$ each are called
 (a) likely (b) equally likely (c) unlikely (d) none of these
4. The probability of a certain event is
 (a) 0 (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1
5. The probability of an event can take the value
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) all of these
6. The probability of an event cannot take the value
 (a) 0 (b) 1 (c) 2 (d) all of these
7. The probability of an impossible event is
 (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) all of these
8. For the following data, find mean, mode and median.

x	5	10	15	20	25	30	35	40
f	2	4	6	8	10	7	5	3

9. Find the median for the following frequency distribution.

Marks	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79
Frequency	3	8	11	5	4

10. The arithmetic mean of 10 values is 35.5. If nine values are 20, 23, 37, 48, 29, 33, 45, 40, 45, find the tenth value

11. Find mean, median and mode for the following frequency distribution.

Classes	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
f	1	5	7	5	1

12. Affan is learning to play dart. He plays it 20 times each day and probability of hitting bull's eye is $\frac{1}{10}$. Find:
 (a) expected number of hits in 25 days. (b) expected number of misses in 25 days.
13. Shifaa usually gets late for her school 2 days out of 6 working days of a week. What is the expected no. of days of her late arrival in 4 weeks.
14. As'ha tries her best to score 100% in each maths assessment, but the probability of getting this is 0.8. Find the expected number of assessments in which she will score 100% out of total 20 assessments.
15. In a lab shelf there are 50 reports of covid 19. Among these 13 are females and 2 of the females are covid positive. If a report is picked at random, find:
 a) The probability of getting a female report
 b) The probability of getting a male report
 c) The probability of getting a female covid negative report.